Srinidhi A
ID-COMETFWC053

EXERCISE - 2.1

The graphs of $y = p(x)$ are given in Fig. 2.10 below, for some polynomials $p(x)$. Find the number of zeroes of $p(x)$, in each case.

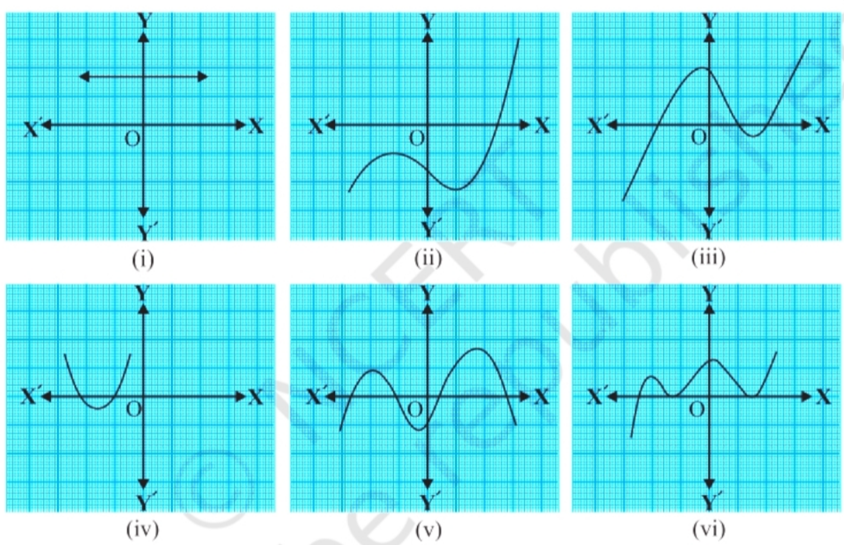

Fig. 2.10

1. **Example 4:** Find a quadratic polynomial, the sum and product of whose zeroes are -3 and 2, respectively.

   Solution: Let the quadratic polynomial be $ax^2 + bx + c$ and its zeroes be α and β. We have

$$\alpha + \beta = -3 = -\frac{b}{a} \tag{1}$$

$$\alpha\beta = 2 = \frac{c}{a} \tag{2}$$

If $a = 1$, then $b = 3$ and $c = 2$. So, one quadratic polynomial which fits the given conditions is $x^2 + 3x + 2$. You can check that any other quadratic polynomial which fits these conditions will be of the form $k(x^2 + 3x + 2)$, where k is a real number. Let us now look at cubic polynomials. Do you think a similar relation holds between the zeroes of a cubic polynomial and its coefficients? Let us consider $p(x) = 2x^3 - 5x^2 - 14x + 8$. You can check that $p(x) = 0$ for $x = 4, -2, \frac{1}{2}$. Since $p(x)$ can have at most three zeroes, these are the zeroes of the given polynomial. Now, the sum of the zeroes is

$$4 + (-2) + \frac{1}{2} = \frac{5}{2} = -\frac{\text{coefficient of } x^2}{\text{coefficient of } x^3} \tag{3}$$

The product of the zeroes is

$$4 \times (-2) \times \frac{1}{2} = -4 = -\frac{\text{constant term}}{\text{coefficient of } x^3} \tag{4}$$

The sum of the products of the zeroes taken two at a time is

$$(4)(-2) + (-2)\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)(4) = -7 = \frac{\text{coefficient of } x}{\text{coefficient of } x^3} \tag{5}$$

In general, if α, β, γ are the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$, then

$$\alpha + \beta + \gamma = -\frac{b}{a} \tag{6}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \tag{7}$$

$$\alpha\beta\gamma = -\frac{d}{a} \tag{8}$$

2. **Example 5*:** Verify that $3, -1, -\frac{1}{3}$ are the zeroes of the cubic polynomial $p(x) = 3x^3 - 5x^2 - 11x - 3$, and verify the relationship between the zeroes and the coefficients.

   Solution: Comparing with $ax^3 + bx^2 + cx + d$, we get $a = 3,\ b = -5,\ c = -11,\ d = -3$. Further,

$$p(3) = 81 - 45 - 33 - 3 = 0 \tag{9}$$

$$p(-1) = -3 - 5 + 11 - 3 = 0 \tag{10}$$

$$p\left(-\frac{1}{3}\right) = -\frac{1}{9} - \frac{5}{9} + \frac{11}{3} - 3 = 0 \tag{11}$$

Therefore, $3, -1, -\frac{1}{3}$ are the zeroes. So, $\alpha = 3, \beta = -1, \gamma = -\frac{1}{3}$. Now,

$$\alpha + \beta + \gamma = \frac{5}{3} = -\frac{b}{a} \tag{12}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = -\frac{11}{3} = \frac{c}{a} \tag{13}$$

$$\alpha\beta\gamma = 1 = -\frac{d}{a} \tag{14}$$

Hence, the relationship is verified.

3. **Example 6:** Divide $2x^2 + 3x + 1$ by $x + 2$.

   Solution: We stop the division process when either the remainder is zero or its degree is less than the degree of the divisor. Here, the quotient is $2x - 1$ and the remainder is 3. Also,

$$(2x - 1)(x + 2) + 3 = 2x^2 + 3x - 2 + 3 \tag{15}$$

$$= 2x^2 + 3x + 1 \tag{16}$$

   That is,

$$2x^2 + 3x + 1 = (x + 2)(2x - 1) + 3 \tag{17}$$

   Therefore, Dividend = Divisor × Quotient + Remainder. Let us now extend this process to divide a polynomial by a quadratic polynomial.